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MTH 105 — Past Questions & Materials

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Course: MTH 105

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Paper 1 of 1 — Coordinate geometry MTH 105

Level 100 • Second Semester • PastQuestion

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UNIVERSITY OF ABUJA
FACULTY OF SCIENCE
DEPARTMENT OF MATHEMATICS
2016/2017 FIRST SEMESTER EXAMINATION

ABU
JAWAL

MTH 105
INSTRUCTION:

TIME:

COORDINATE GEOMETRY UNIT: 1
ANSWER ALL QUESTIONS IN SECTION A
AND ANY ONE (1) QUESTION IN SECTION B
1 HOUR, 30 MINUTES

1. A circle has the equation $x^2 + y^2 = 16$. What is the radius of this circle?
(a) 256 (b) 8 (c) 16 (d) 4
2. A circle has the equation $(x + 3)^2 + (y - 1)^2 = 4$. Which of the following statements is **false**?
(a) The x coordinate of the centre is -3 (b) The radius of the circle is 2
(c) The y coordinate of the centre is -1 (d) The point (-3, -1) lies on the circle
3. O is the centre of a circle. P is a point on the circumference. The gradient of OP is 2.
What is the gradient of the tangent at P?
(a) 2 (b) 0.5 (c) -2 (d) -0.5
4. Determine the x-intercept and the y-intercept for the graph of the equation: $2x - 3y + 24 = 0$
a. x-intercept: -12; y-intercept: -8 b. x-intercept: 12; y-intercept: 8
c. x-intercept: -12; y-intercept: 8 d. x-intercept: 12; y-intercept: -8
5. A line has x-intercept 4 and y-intercept 7? Determine the slope of the line.
(a) $\frac{7}{4}$ (b) $-\frac{4}{7}$ (c) $-\frac{7}{4}$ (d) $\frac{4}{7}$
6. Write an equation for the line that passes through T(-1, 4) and is parallel to the line $y = 6x - 6$.
(a) $y + 4 = \frac{-1}{6}(x - 1)$ (b) $y + 4 = 6(x - 1)$ (c) $y - 4 = \frac{-1}{6}(x + 1)$
(d) $y - 4 = 6(x + 1)$
7. The slope of line L is $\frac{-1}{3}$. What is an equation of a line that is perpendicular to line L?
(a) $y + 2 = \frac{1}{3x}$ (b) $-2x + 6 = 6y$ (c) $9x - 3y = 27$ (d) $3x + y = 0$
8. What is the equation of a circle whose center is 4 units above the origin in the coordinate plane and whose radius is 6?
(a) $x^2 + (y - 6)^2 = 16$ (b) $y^2 + (x - 6)^2 = 16$ (c) $x^2 + (y - 4)^2 = 36$
(d) $y^2 + (x - 4)^2 = 36$
9. What is the slope of a line that is perpendicular to the line whose equation is $3x + 5y = 4$?
(a) $-\frac{3}{5}$ (b) $\frac{3}{5}$ (c) $\frac{5}{-3}$ (d) $\frac{5}{3}$
10. A line segment has endpoints A(7, -1) and B(-3, 3). What are the coordinates of the midpoint of $\overline{AB}$?
(a) (1, 2) (b) (2, 1) (c) (-5, 2) (d) (5, -2)
11. In circle O, a diameter has endpoints (-5, 4) and (3, -6). What is the length of the diameter?
(a) $\sqrt{2}$ (b) $2\sqrt{2}$ (c) $\sqrt{10}$ (d) $2\sqrt{41}$

12. A circle has the equation $(x - 2)^2 + (y + 3)^2 = 36$. What are the coordinates of its center and the length of its radius?
 (a) $(-2, 3)$, 6 (b) $(2, -3)$, 6 (c) $(-2, 3)$, 36 (d) $(2, -3)$, 36
13. Find the equation of the straight line which passes through $(-2, 3)$ and $(4, -5)$
 (a) $3y + 4x - 1 = 0$ (b) $4y + 3x - 1 = 0$
 (c) $3y - 2x - 1 = 0$ (d) $-5y + 4x - 1 = 0$
14. The equations of $\overline{PQ}$ and $\overline{PR}$ are $x - 3y + 11 = 0$ and $3x - y + 1 = 0$ respectively. Find the coordinates of P.
 (a) $(1, 4)$ (b) $(-1, 4)$ (c) $(4, -1)$ (d) $(-4, -1)$
15. Find the acute angle between $3x - y + 1 = 0$ and $x - 2y + 1 = 0$
 (a) 32.2° (b) 52.2° (c) 18.2° (d) 28.2°
16. Find the equation of the tangent to the circle $x^2 + y^2 = 13$ at the point $(2, -3)$
 (a) $2x - 3y = 13$ (b) $3x - 2y = 13$ (c) $x - 3y = 13$ (d) $2x - y = 13$
17. Find the point at which the line $3y - 12 = 4x$ cuts the y-axis.
 (a) $(0, 3)$ (b) $(0, 4)$ (c) $(0, 12)$ (d) $(0, 0)$
18. The two lines represented by the equations $x + y = \frac{2}{3}$ and $3(x - 2) = -3y - 4$ are graphed on a coordinate plane. Which statement best describes the two lines?
 (a) The lines are parallel.
 (b) The lines are the same line.
 (c) The lines are perpendicular.
 (d) The lines intersect at an angle other than 90° .
19. $(7, -2)$ and $B(-3, 3)$ are the end points of the line $\overline{PQ}$. Find the coordinate of the point A which divides the line the ratio 3:2.
 (a) $(1, -1)$ (b) $(1, 0)$ (c) $(0, 1)$ (d) $(1, 1)$
20. Determine the value of k if the equations $4kx - 2y - 10 = 0$ and $3x + 5y - 7 = 0$ represent parallel lines.
 (a) $\frac{-3}{7}$ (b) $\frac{-3}{10}$ (c) $\frac{-4}{7}$ (d) $\frac{-5}{7}$

SECTION B

1. The straight line $3x - 4y - 3 = 0$ cuts the circle $x^2 + y^2 - 16x - 8y + 30 = 0$ at the points P and Q
 a. Find the coordinates of P and Q
 b. Calculate the length of the chord PQ.
2. Let the lines L, M and N be represented by the equations $x + y - 5 = 0$, $2x + 3y - 13 = 0$ and $5x - 2y - 4 = 0$, respectively.
 a. Show that the lines L, M and N are concurrent.
 b. Find their common point